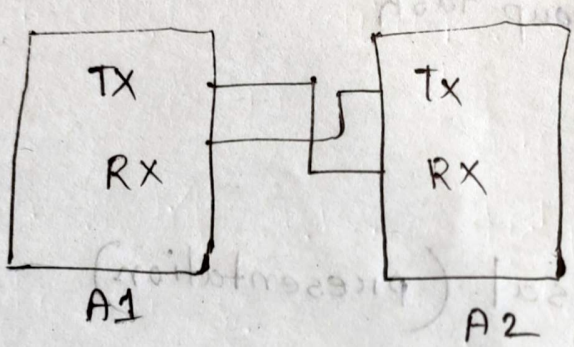
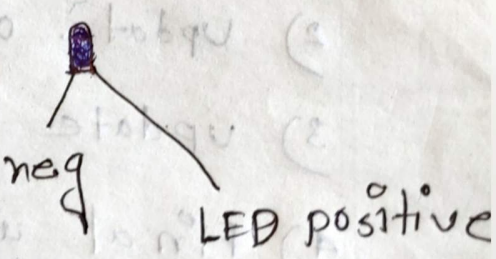
# Lab - 1

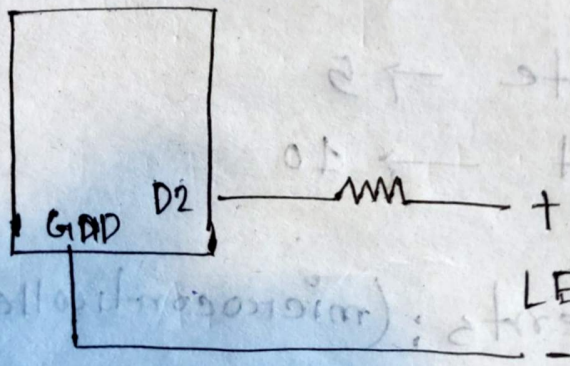
- Digital Pin → value (0/1)
  ↓
  Digital sensor connect করতে হবে
  (analog ও ও করা যায় না)

Tx – Transmit
RX – Receive




Digital LED:




- setup( ) : 1 time execute,
  initialization (Hardware)
  pin – input / output / config
  Data transfer rate.

Board — Arduino UNO

Port — COM6 ↓ connect

loop ( ) : Repeatedly work.

For light blink:

```
void setup( ) {
    pinMode ( 2, OUTPUT ); // pin configure করা
}

void loop ( ) {
  digitalWrite (2, HIGH);
    delay (1000);
    digitalWrite (2, LOW);
    delay (1000);
}
```

pin no → 2, 5V → HIGH, 0V → LOW, ms (1 s) → delay (1000)

Ard. UNO LED → 2 NO pin → LED তে output দিচ্ছে

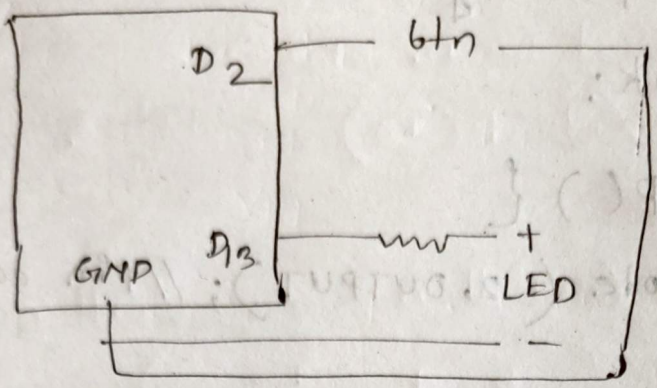
# digitalWrite ( , H)
digitalWrite ( , L)
delay (1000);

} output off করবো

# digitalwrite ( , H)
digitalwrite ( , L)

} vary person to person.

with button:




#
pinMode(2, INPUT_PULLUP)
↓
5V pin থেকে কারেন্ট চলে আসে।

btn not press → 2 No pin → 5V
btn press → 2 No pin → 0V
[Vol Ground এ চলে যায়]

btn press না করলে (false) → else এ যাবে। means light off

```
if (sensorVal == LOW)
  { digitalWrite(13, HIGH);
  }
else { digitalWrite(13, LOW);
  }
```

btn press করলে, if true LED জ্বলবে

3/4 ☑ light এর value বেশি হলে LED on
light " " কম " LED off

# Lab-2

Gas sensor

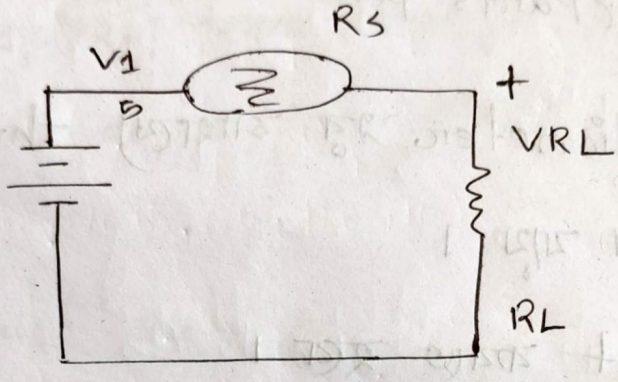
AO → Analog output (value)

DO → Digital output (0/1)

↳ Limit << threeshold (1)
Limit >> -- (0)

GND → GND

Vcc → 5V




→ Gas sensor

Gas volume ↑ Rs ↑

Rs ↑ voltage Drop ↑

Arduino : A D C

Analog Digital converter

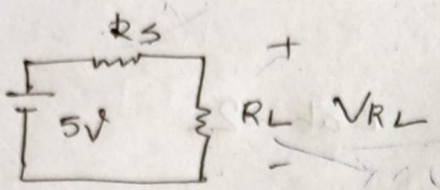
ADC → 10 bit → $2^{10}$ No (0 - 1023)

5v → 1023 (heighest value)

∴ 1v → $\frac{1023}{5}$

∴ xv drop → $\frac{1023 \times x}{5} = y$ [sensor value]

code: VRL → $x = \frac{5y}{1023}$




Mid Final ☐ $V_{RL} = \frac{R_L}{R_S + R_L} \times 5$

$\Rightarrow R_S = \frac{5R_L}{V_{RL}} - R_L$ ] Rs result

ppm → Parts per million

☐ Potentiometer এর সাহায্যে threshold মান/ রেঞ্জ করা যায়।

☐ Preheat করতে হবে।

Digital (Display) [OLED → Arduino]

I2C = In

SCL — A5

SDA — A4

GND — GND

VCC — 5V

⊞ display . begin → মুছবে

for ( ;; ) → infinite loop

Lab Report (2) :

1. class এর ২টা বাচ্চা merge করবে (sensor এর reading এর যে data থাকছে সেটার sensor এর সেখাবে ) .

2. buzzer

3. - - - just lekha (সেখাবে

4. X বাদ

5. exhast fan (LED use করবো)

Lab-1 :

$$\text{Distance} = \frac{\text{Time}}{x}$$

ARDUINO UNO:

digital pin → 5v (high) } digital input
→ 0v (low) } or output.

Analog pin → $A_0$ to $A_5$ (Analog input)

delay( ) → pauses the program.

setup( ) → Executes once for initialization

loop( ) → Executes repeatedly to run the main program.

## Arduino :

Arduino is an open-source hardware and software platform used to build electronic and embedded system projects

* Arduino board → microcontroller board

↓

(i) Microcontroller
(ii) Input pins
(iii) output pins
(iv) USB programming Interface.

Arduino Board — (i) Arduino UNO ✓
(ii) " NANO
(iii) " Mega

* We use UNO because it is sufficient for mid-level hardware projects.

Application of Arduino UNO:

(i) Home Automation
(ii) Robotics
(iii) IOT
(iv) Environmental Monitoring
(v) security system
(vi) Weather Monitoring
(vii) Smart Agriculture
(viii) Healthcare Monitoring

⊞ RX এবং TX :

serial communication (UART)

computer এর সাথে data আদান - প্রদান

করে।

⊞ Arduino IDE:

Arduino IDE is the software used to write, compile and upload Arduino Program

LED Toggling : বার বার light on off.

Arduino ADC:

Analog voltage কে digital voltage এ রূপান্তর করে।

Arduino সরাসরি Analog voltage বুঝতে পারে না।

Ex: bit = 12

ADC, value $= 2^{12} = 4096$

Range = (0 – 4095)

5 V → 4095

∴ 1 V → $\frac{4095}{5}$

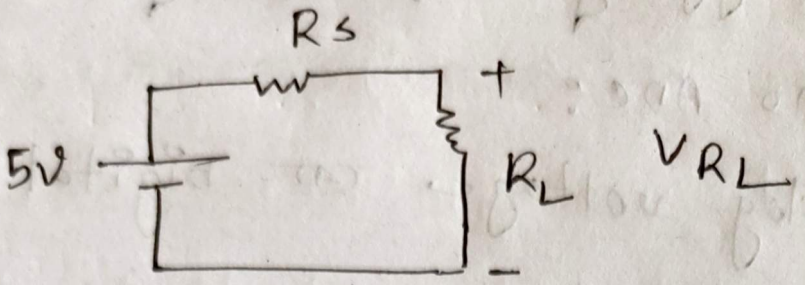
∴ x V → $\frac{4095 \times x}{5} = y$

Normal voltage

ADC voltage

∴ VRL → $x = \frac{5y}{4095}$

Gas sensor:




$$V_{RL} = \frac{R_L}{R_S + R_L} \times 5$$

$$\Rightarrow R_S = \frac{5R_L}{V_{RL}} - R_L$$

↓

sensor Resistance

Gas sensor preheat করার জন্য:

```
void setup( ) {
  Serial.begin(9600);
  Serial.println("MQ2 warming up!");
  delay(20000);
}
```

↓

preheat হচ্ছে।

## ESP32 :

ESP32 is a system on a chip microcontroller with built-in WiFi and Bluetooth mainly used for developing IOT applications.

ESP32 কেন IOT এর জন্য ব্যবহৃত করা হয়:

1. Built-in WiFi
2. Built in Bluetooth
3. Fast Processor
4. GPIO Pins
5. ADC
6. DAC
7. Low power consumption.

- 18 ADC channels
- 3 UART interfaces.
- 520 KB RAM.

DHT11 sensor:

DHT11 হলো Temperature and Humidity sensor.

XAMPP ব্যবহার —

(i) Local server তৈরি করতে —

(ii) website Test করতে —

(iii) Database ব্যবহার করতে —

(iv) PHP program চালাতে —

(v) ESP32 থেকে Data গ্রহণ করতে

---

(1) Apache — web server হিসেবে কাজ করে।

Browser — এর request গ্রহণ করে

and web page দেখায়।

(2) MySQL: Database এ data store করে।

(3) PHP : Database থেকে Data পড়ে

এ এ Data save করে

Dynamic web page তৈরি করে।

#include <WiFi.h>

↳ used to connect the ESP32 to a WiFi Network.

WL_CONNECTED → ESP32 successfully connected.

I2C communication:

SDA, SCL connect করতে হবে

Serial.begin(9600) : ~~Arduino UNO connec~~

initializes serial communication between the UNO and computer at a baud rate of 9600 bits per second.

#

```
digitalWrite(1, High)
delay(3000)
digitalWrite(1, LOW)
delay(2000)
```

delay() is used to pause the execution of the program for a specified amount of time

Input Device: Sends data / signals to UNO

Ex: Push button / MQ2 Gas / LDR

Output Device: Recieves signal from Arduino and produces an output

Ex: LED, OLED, buzzer

PINmode() is used to configure a pin as INPUT or OUTPUT

DHT11 - Temperature and Humidity

MQ2 - Gas & smoke

GPIO - General Purpose Input Output

↳ GPIO pins are used to connect sensors and output devices. They can be configured as input or output pins and can also support communication protocols such as UART, I2C, SPI and PWM.

| Arduino | ESP32 |
|---|---|
| 1. No Built in WIFI or bluetooth | 1. YES |
| 2. Less Processing power (Slow) | 2. More (Faster) |
| 3. Best for basic projects | 3. Best for IOT and wireless project |

| Digital Read() | Analog (Read) |
| --- | --- |
| Reads High or LOW | Reads Analog Value |
| 0 or 1 | 0 - 1023 |
| Push Button, Switch | MQ2, LDR, Potentiometer |
| | A0 - A5 |

ESP32 projects:

(i) Smart Home Automation

(ii) Wireless IOT monitoring system

(iii) Gas Leakage Detection System

(iv) Smart parking system

Serial.println(): Its used to print data or messages on the Serial Monitor

Digital Write(): is used to set a digital output pin to either HIGH or Low. It is used to control output devices such as LEDs, buzzers and relays.

---

ESP 32 — DHT11

D19 ——— SIG

3.3V ——— VCC

GND ——— GND

↙ digital

---

Arduino — MQ-2 Gas Senso

Analog

5V ——— VCC

GND ——— GND

2 (digital Pin) ——— DO

A0 ——— A0

| Arduino | OLED |
|---|---|
| 5V | VCC |
| GND | GND |
| A4 | SCL |
| A5 | SDA |

Buzzer + Push btn:

| Push btn | Arduino |
|---|---|
| One leg | → A2 |
| other leg | → GND |

| Buzzer | |
|---|---|
| (+) | → A8 |
| (-) | → GND |

| Arduino | — | HC - SR04 (sonar sensor) |
|---|---|---|
| 5V | — | VCC |
| GND | — | GND |
| D9 | — | TRIG |
| D10 | — | ECHO |

# Rasberry Pi

Microprocessor : i) consumes more power

ii) operates at 700 MHz - 1.5 GHz

iii) RAM size - (1-8) GB

Rasberry pi can run both 32-bit and 64-bit programs.

- R-Pi need microSD card - capacity atleast 8GB.
- R-Pi has 2 micro-HDMI ports, to connect 2 separate monitors.
- DVI Port → using HDMI to DVI cable.
- VGA Port → VGA to HDMI adapter.
- Image Rotation possible → 90°, 80°, 270°

To reset the image, set rotation to 0.

- add a loop to take five picture in a row:

```
camera.start_preview()
    for i in ranges(5):
        sleep(5)
        camera.capture('1ink')
camera.stop_preview()
```